

Some notes about Microsoft Exchange Deserialization RCE (CVE-2021-42321)

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Exchange

Tianfu

2021

Rce

Deserialization

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[\[image: Peterjson\]](#)

](https://peterjson.medium.com/?source=post_page---byline--110d04e8852-----
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[Peterjson](#)

8 min readNov 19, 2021

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actionUrl=https%3A%2F%2Fmedium.com%2F_%2Fvote%2Fp%2F110d04e8852&operation
=register&redirect=https%3A%2F%2Fpeterjson.medium.com%2Fsome-notes-about-
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Vietnamese version: <https://testbnull.medium.com/some-notes-of-microsoft-exchange-deserialization-rce-cve-2021-42321-f6750243cdcd>

INTRO

It's been several months since our last story about ****ProxyShell Exploit**** and recently Exchange was pwned again at Tianfu Cup 2021. We're very excited about that Exploit and we're waiting for Tuesday Patch of MS Exchange this month to analyse it.

There's already a [blog](#) analysis about CVE-2021-42321 and just published yesterday. But we think that blog didn't cover enough technical information and highlight notes so we decided to write this blog in English to let everyone understand what happened inside this CVE!

From the advisory of Microsoft, it stated that this CVE is a post-auth RCE. We just wonder that is a pre-auth RCE because it costs \$200.000 when you have a successful demonstration at Tianfu Cup 2021. But with the patch from MS we only know that MS patch the post-auth RCE, maybe MS let the customer have time to patch the post-auth RCE and later release another patch for an auth bypass vulnerability?

<https://msrc.microsoft.com/update-guide/vulnerability/CVE-2021-42321>

If we look carefully at the advisory of Microsoft we can notice that only Exchange 2016 CU 21,22 and Exchange 2019 CU 10,11 . This means the only recent latest version of Exchange 2016,2019 are vulnerable to this CVE

<https://msrc.microsoft.com/update-guide/vulnerability/CVE-2021-42321>

Microsoft also release a patch for Exchange 2016,2019 before the Tianfu Cup happened and Exchange was pwned after this patch, so we need to diff the patch October and November 2021.

<https://www.catalog.update.microsoft.com/Search.aspx?q=Exchange%202016>

After decompiling with DnSpy and diffing with Win Merge we got about 275 files were changed (J°□°□□ L).

The patch almost patching for Deserialization Sink inside Exchange, so we can be sure that this time it's a deserialization vulnerability.

THE SINK

We noticed that some files were removed

There's a funny point inside **TypedBinaryFormatter** is when deserialize the **binder* ***value was not passed correctly to *ExchangeBinaryFormatterFactory **. As we already know that ***SerializationBinder *** is used to control the actual types used during serialization and deserialization. This can be a mechanism deserialization vulnerability but in ***TypedBinaryFormatter *** we have a chance for it again :) .

Microsoft.Exchange.Compliance.Serialization.Formatters.TypedBinaryFormatter

Without the **binder** **variable pass to*

*****ExchangeBinaryFormatterFactory.CreateBinaryFormatter() ****,
*****ExchangeBinaryFormatterFactory ***** will use *****ChainedSerializationBinder ***** as
*****SerializationBinder ***** to validate ***actual types used during deserialization*

Microsoft.Exchange.Diagnostics.ExchangeBinaryFormatterFactory.CreateBinaryFormatter()

When passing the argument from **TypedBinaryFormatter.Deserialize()** to **ExchangeBinaryFormatterFactory.CreateBinaryFormatter()** **Exchange initialize*
*****ChainedSerializationBinder *(implement SerializationBinder) ***** with:

- strictMode = **false**
- allowList = System.DelegateSerializationHolder
- allowedGenerics = null

Let's have a look at **ChainedSerializationBinder.BindToType()** which will validate the class for Deserialization.

ChainedSerializationBinder.BindToType()

ChainedSerializationBinder.ValidateTypeToDeserialize()

For Block 1:

- If the strictMode is **False** ****and our class is not in allow list, and in blacklist , Exchange will throw **InvalidOperationException**
- In this function, it only catches **BlockedDeserializationException** ****so** **InvalidOperationException** ****will not be caught, so if Exchange through** ****InvalidOperationException** ****our deserialization chain will be broken**

For Block 2:

- If there's a **BlockedDeserializationException** ****while ValidateTypeToDeserialize() was thrown, it will be caught in catch block but Exchange only throws that Exception when the value of the flag is True, but remembered that the strictMode value was passed to **ChainedSerializationBinder **is always False ! So there's Exchange will catch **BlockedDeserializationException** safely and Exchange didn't affect our Deserialization can continue without crashing.

Alright, now let's have a look at ****ChainedSerializationBinder's **blacklist**. It's a big mistake of Exchange developers ...

When deserialize ClaimsPrincipal, it also triggers another **BinaryFormatter.Deserialize()** without any filter. At this time, we can be sure that we found the Sink of this CVE

```
System.Security.Claims.ClaimsPrincipal.OnDeserializedMethod()
    System.Security.Claims.ClaimsPrincipal.DeserializeIdentities()
        BinaryFormatter.Deserialize()
```

THE SOURCE

With DnsSpy we tracing back to find where we can trigger deserialization

After some handy steps we found that, we can go from **OrgExtensionSerializer.TryDeserialize()** to **ClientExtensionCollectionFormatter.Deserialize()**

It gets the Stream from userConfiguration to deserialize it and sound like with a normal user we can set that setting also.

After a little bit of playing around with how can we set the ****userConfiguration** ****from HTTP request we found this document from Microsoft**

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CreateUserConfiguration operation

The CreateUserConfiguration operation creates a user configuration object on a folder. The following example of a...

docs.microsoft.com

](https://docs.microsoft.com/en-us/exchange/client-developer/web-service-reference/createuserconfiguration-operation?source=post_page-----110d04e8852-----))

But from the public document, we can't find how to set Stream into userConfiguration so we dive into the logic of EWS to see how we can set it

We clearly see that there's a XML node name "BinaryData", which sound like the serialization data we want to set

```
**CreateUserConfiguration**.*Execute()  
    ***UserConfigurationCommandBase**.*SetProperties()  
        ***UserConfigurationCommandBase**.*SetStream()*
```

UserConfigurationCommandBase.*SetProperties()*

UserConfigurationCommandBase.*SetStream()*

Basically, our request to EWS looks like this

sample request for setting BinaryData

Now we can set ****userConfiguration**** but how to trigger the actual deserialization sink?

Tracing back from **OrgExtensionSerializer.*TryDeserialize()** ****we can reach to *GetClientAccessToken**

With this API from MS documents, now we can trigger the deserialization sink

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GetClientAccessToken operation

Find information about the GetClientAccessToken EWS operation.
The GetClientAccessToken operation gets a client access...

docs.microsoft.com

](https://docs.microsoft.com/en-us/exchange/client-developer/web-service-reference/getclientaccesstoken-operation?source=post_page-----110d04e8852-----))

*sample request for **GetClientAccessToken***

FULL EXPLOIT

We finally achieve post-auth RCE:

- Create UserConfiguration with BinaryData as our Gadget Chain
- Request to EWS for GetClientAccessToken to trigger the Deserialization

About the gadget chain:

- We can embed any ysoserial.net gadget chain into `**System.Security.Claims.ClaimsPrincipal **` to trigger 2nd Deserialization
- Or we can use directly use `**TypeConfusedDelegate **Chain` (This chain is not in the blacklist of `**ChainedSerializationBinder **` and I don't know why :>)

IMPROVEMENT

We already can pop calc on our Lab environment but how about the actual environment with up to date Windows Defender?

This is what we got :)

Since ProxyLogon, ProxyShell, and till now some EDRs, AV, sysmon and Microsoft Windows Defender try to catch and prevent process spawn from w3wp.exe process. This also annoys us but we need some improvements to overcome it!

As we know that we can use **ClaimsPrincipal** `**or **TypeConfusedDelegate` gadget chain to achieve post-auth RCE, but directly executing cmd.exe is not a good idea

With some improvements from @zcgonvh to ysoserial.net , gadget

ActivitySurrogateSelector can use to load a DLL via `** Assembly.Load() **` function. The whole idea is from @zcgonvh blog also.

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Home of the Alpacas - analysis and weaponization of the CVE-2020-17144 vulnerability

0x00 Foreword Similar to CVE-2018-8302 and CVE-2020-0688, CVE-2020-17144 is also a deserialization vulnerability that requires being logged in to exploit, but it only affects Exchange 2010 servers. ...

www.zcgonvh.com

](http://www.zcgonvh.com/post/analysis_of_CVE-2020-17144_and_to_weaponizing.html?source=post_page-----110d04e8852-----)

This helps us achieve mem-shell when exploiting .net deserialization. Instead of calling `Process.Start()` now we move on `eval JScript`. With `JScript`, we can do many things with scripting instead of spawning `cmd.exe / powershell.exe`.

A snippet of code to eval JScript

Next, we write a new plugin for ysoserial.net with **ClaimsPrincipal** `**(this is almost the same with ClaimsIdentity)` then put `**ActivitySurrogateSelector **` object into `ClaimsPrincipal **` gadget chain.

But `**ActivitySurrogateSelector **` has some limitations with .NET 4.8 and later

Shout out to [@monoxgas](#) for **ActivitySurrogateDisableTypeCheck** gadget chain to disable the protection for **ActivitySurrogateSelector**

Everything we need now is to put **ActivitySurrogateDisableTypeCheck** **into ClaimsPrincipal**

We've been laughing for hours while making this meme

The main idea of **ActivitySurrogateDisableTypeCheck** gadget chain is when deserializing it will call *GetObjectFromSerializationInfo()* and finally reach to *XamlReader.Parse()* and run our Xaml payload to set value for

DisableActivitySurrogateSelectorTypeCheck

Our Xaml payload was executed but ...

There's an **InvalidCastException** ****was thrown before we can set**

****DisableActivitySurrogateSelectorTypeCheck** to True, and this Exception was not caught by Exchange so this chain won't work and we cannot change

****DisableActivitySurrogateSelectorTypeCheck** ****because of crashing w3wp.exe process!**

So we can't use **ActivitySurrogateDisableTypeCheck** ****gadget chain. Now we need to find another gadget chain or another way to call **XamlReader.Parse()**

Remember that we can use **TypeConfusedDelegate* *?*

With ****TypeConfusedDelegate** ****we can invoke the method arbitrarily (◡‿◡)**

This is how we call Process.Start()

How about **XamlReader.Parse()** ? **XamlReader.Parse** ****is a public/static also, so we can easily call it with **TypeConfusedDelegate**

from TypeConfusedDelegate -> XamlReader.Parse()

Finally, everything works as our expected, we can change

****DisableActivitySurrogateSelectorTypeCheck** ****to True to overcome the limitation of .NET and later inject DLL to achieve mem-shell with Jscript to bypass the detection**

PoC video:

PoC: We don't think we will ...

Final Thought

This exploit only work for

- Microsoft Exchange 2019 CU10, 11
- Microsoft Exchange 2016 CU21, 22

And didn't affect another version, who knows what happened behind the scenes with a typo mistake on **ChainedSerializationBinder **`^(\[)/`